

TECHNICAL PROJECT REPORT

Project Title: NAVISECURE: An IoT-Based Truck Driver Safety & Monitoring System

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1. Executive Summary

In the modern logistics and transport sector, long-haul multi-axle trucks form the backbone of interstate commerce. However, while consumer passenger vehicles have seen rapid advancements in Advanced Driver Assistance Systems (ADAS) and proactive safety features, heavy commercial vehicles remain technologically outdated. This gap in safety technology leads to high accident rates, primarily caused by driver fatigue, unauthorized operation, mobile phone distractions, and delayed emergency responses due to remote crash locations.

Navisecure is an integrated IoT-based hardware and software prototype designed to safeguard truck drivers, fleet assets, and cargo. The system introduces a secure, multi-layered defense architecture:

1. **Pre-Drive Security:** Iris scanning biometric authentication and secure smartphone pairing, paired with an automated pre-drive checklist that prevents ignition if critical checks fail.
2. **In-Transit Distraction Control:** A smart call management module restricting mobile phone distractions to a pre-approved list of priority contacts with strict time-outs (25-30secs).
3. **Post-Incident Response:** An automatic crash detection system using Inertial Measurement Units (IMUs) that broadcasts GPS coordinates and triggers automated emergency calls via GSM within 60 seconds of a collision.
4. **Data Forensics & Monitoring:** A robust Black Box logging unit and cloud-connected backend enabling real-time fleet oversight and accident investigation capabilities.

By shifting safety paradigms from passive protection to proactive, connected intervention, Navisecure delivers a scalable, life-saving platform for the logistics industry.

2. Introduction & Problem Statement

2.1 Context of Heavy Vehicle Logistics

Long-haul truck drivers operate under demanding conditions, frequently driving overnight for extended hours with minimal supervision. Because these vehicles travel on high-speed national highways and interstate routes carrying massive payloads, any lapse in driver concentration or mechanical failure has catastrophic consequences.

2.2 The Core Problem

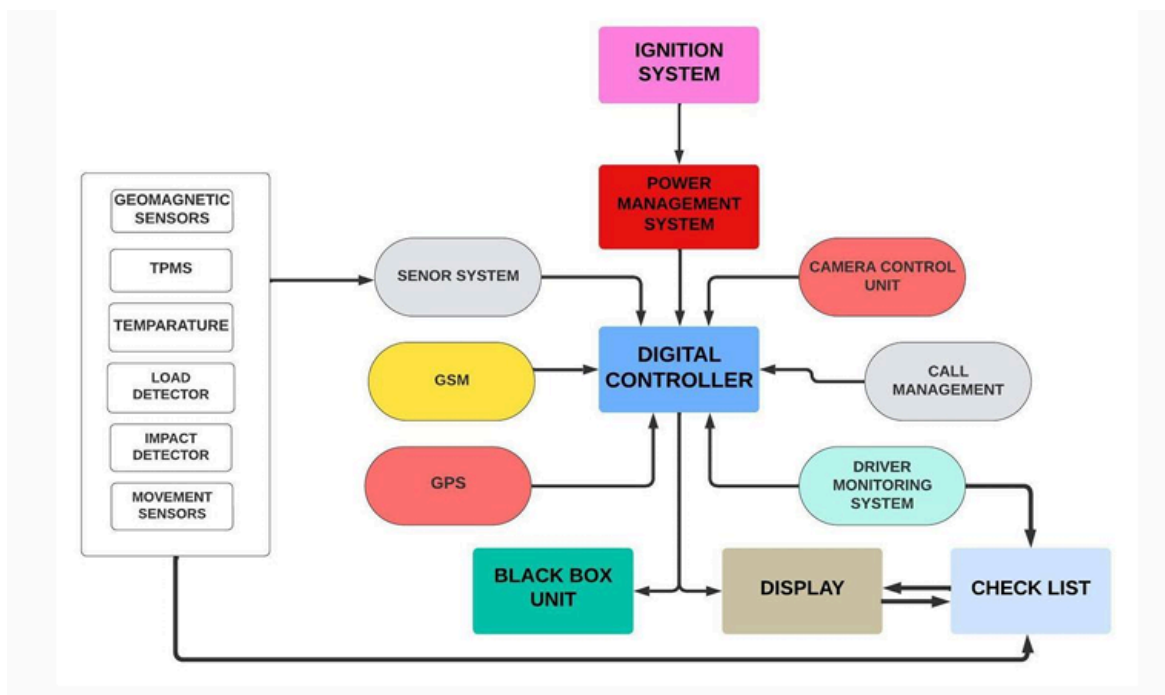
Despite representing a fraction of total registered vehicles, heavy commercial trucks are disproportionately involved in fatal road accidents. The core vulnerabilities can be classified into four primary domains:

1. **Lack of Driver & Vehicle Monitoring:** No systemic measures exist to verify if the driver behind the wheel is authorized, rested, or if the truck is mechanically fit to commence a trip.
2. **Mobile Phone Distraction:** Incoming calls and continuous smartphone interaction during highway driving degrade driver focus, significantly increasing the risk of lane departure and rear-end collisions.
3. **Delayed Emergency Response:** When a crash occurs in remote or rural locations, victims are often incapacitated. The inability to communicate immediate, precise coordinates (x,y GPS coordinates) leads to prolonged rescue windows, directly impacting survivability.
4. **Absence of Diagnostic Data:** Fleet managers remain blind to real-time anomalies (e.g., severe tire pressure drop, cabin overheating, overloads) until a breakdown occurs on the road.
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3. System Architecture & Block Diagram Analysis

The Navisecure architecture is centered around a centralized **Digital Controller** that processes inputs from various sensor arrays and coordinates actions across the ignition, communication, and human-machine interface (HMI) modules.

3.1 Flow and Block Diagram Representation

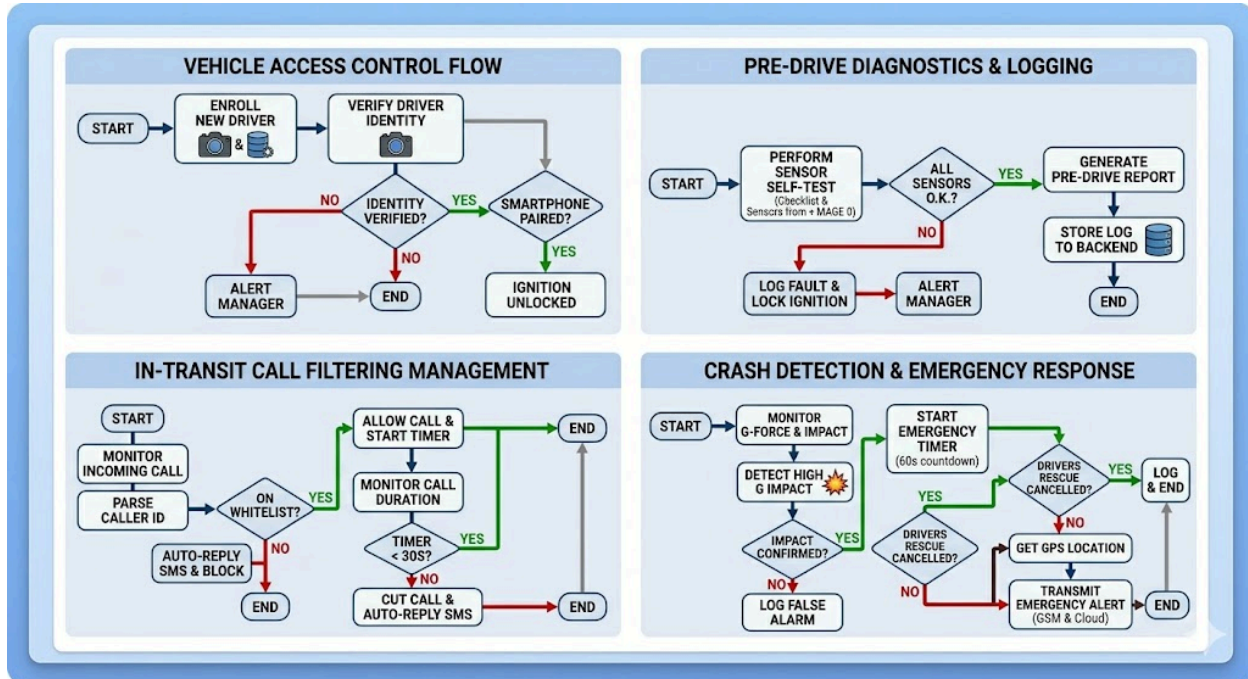


3.2 Detailed Module Analysis

- **Digital Controller (Central Processing Unit):** An IoT-capable, high-performance microcontroller (e.g., ESP32 or ARM Cortex-M based board) that handles real-time sensor polling, runs local state-machine logic, and interfaces with the communication modules.
- **Sensor System Sub-Array:**
 - *Geomagnetic Sensors:* Used for secondary orientation and compass heading calculations to support GPS trajectory verification.
 - *Tire Pressure Monitoring System (TPMS):* Constantly monitors tire pressure and cavity temperature to prevent blowouts.
 - *Temperature Sensors:* Monitors the cabin climate and engine compartment to prevent overheating incidents.
 - *Load Detector:* Verifies that cargo weight matches legal operating capacities, preventing commercial overloading.
 - *Impact Detector (Accelerometer/Gyroscope):* High-G triaxial accelerometer (>8G sensing threshold) designed to register sudden, severe deceleration vectors indicative of a collision.
 - *Movement Sensors (PIR/Ultrasonic):* Detects physical driver presence and posture within the cabin workspace.
- **Power Management System (PMS):** Isolates the sensitive microelectronics from the noisy 24V DC environment of heavy-duty trucks, providing clean 5V and 3.3V lines with built-in surge and reverse-polarity protection.
- **Driver Monitoring System (Biometrics & Camera):** Interfaces with an IR-based iris scanning camera to perform fast, local biometric verification.
- **Display / Human-Machine Interface (HMI):** A dashboard-mounted LCD panel that guides the driver through verification, displays the pre-drive checklist, and presents real-time diagnostic warnings.
- **Communication Subsystem (GSM & GPS):**
 - *GPS Module:* Constantly tracks latitude, longitude, speed, and altitude.
 - *GSM/GPRS Module:* Establishes standard cellular connectivity for sending emergency SMS telemetry, placing emergency voice calls, and synchronizing system logs with the fleet server.
- **Black Box Unit:** Secure, onboard non-volatile memory (EEPROM / industrial micro-SD) that records sensor data in a circular buffer format, preserving the final 10 minutes of driving telemetry for post-accident crash forensics.

4. System Implementation Methodology

The execution of the Navisecure system is organized into distinct functional phases, guiding the system from physical verification to real-time mitigation and emergency response.



Phase 1: Driver Verification & Secure Ignition Handshake

1. Iris Scanning Protocol:

- When the cabin door opens, the onboard IR camera wakes up and prompts the driver for an iris scan via the dashboard LCD.
- The system compares the captured pattern with templates stored securely in local secure flash storage (populated during driver onboarding).

2. Smartphone IoT Handshake:

- Simultaneously, the driver's smartphone (running the dedicated Navisecure App, which is in process of building an application) searches for the truck's localized Bluetooth Low Energy (BLE) or Wi-Fi hotspot.
- Upon detection, a cryptographic handshake occurs. Both factors (Verified Iris + Paired Smartphone) must return TRUE to authorize the next stage.

Phase 2: Pre-Drive Safety Checklist

Once verification is complete, the dashboard screen displays a mandatory interactive checklist.

- The driver must manually confirm tire status, visual mirror alignments, and seatbelt attachment.
- The system cross-checks statements with onboard sensors
- Ignition Lock:** If any critical metric (e.g., low tire pressure or unverified driver) fails, the

Digital Controller maintains an open relay on the vehicle's ignition starter line, physically preventing the engine from starting.

Phase 3: In-Transit Smart Call Management

When the vehicle is in motion (indicated by GPS speed >5km/hr or transmission sensors), the paired smartphone app enters "Driving Mode":

- **Call Filtering:** The app suppresses all incoming calls, with the exception of **5 pre-registered priority contacts** (e.g., Fleet Manager, Immediate Family, Logistics Dispatch).
- **Call Duration Control:** Incoming calls from these priority contacts are capped at a strict threshold of 25-30 seconds. A visual and auditory warning sounds at 20 seconds to notify the driver.
- **Conversation Override Lock:** If an urgent conversation must extend beyond 30 Seconds, the system enforces safety through state-awareness: the driver must park, turn off the engine, and decouple from the system before call restrictions are lifted.
- **Automated SMS Engine:** For any calls blocked during transit, the app immediately dispatch a templated auto-reply SMS: *"The driver is currently on the road and will return your call later."*

Phase 4: Collision Detection & Automated Emergency Telemetry

The system utilizes continuous polling of the triaxial accelerometer to detect severe impacts.

$$\text{Deceleration Vector } (G) = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

- **Trigger Condition:** If the calculated deceleration force exceeds a pre-defined safety threshold (e.g., $G \geq 4.5G$), an emergency state is initiated.
- **60-Second Recovery/Cancel Grace Window:** A 60-second countdown is displayed on the dashboard with a loud audio prompt. If it was a false positive or minor scrape, the driver can enter a secure bypass PIN on the screen to cancel the dispatch.
- **Automated Dispatch Protocol:** If the timer expires without cancellation, the system performs two simultaneous actions within under 60 seconds:
 1. **Telemetry SMS:** Sends an emergency text message to the pre-configured fleet dispatch and local emergency services containing the truck's registration number, current GPS coordinates, speed prior to impact, and diagnostic status.
 2. **Emergency GSM Call:** Initiates an automated voice call to the emergency contact, establishing a direct cabin speakerphone line so responders can listen for signs of driver consciousness.

5. Technical Specifications & Bill of Materials (BOM)

To implement the Navisecure system, the following hardware and software stack is utilized:

5.1 Hardware Component Specifications

Component Category	Part / Model Suggestion	Key Functionality & Parameters
Central Processor	ESP32-WROOM-32D / ARM STM32	Dual-core processing, built-in Wi-Fi and BLE, multi-channel ADC.
Biometric Module	PixArt/CMOS Iris Sensor	Near-Infrared (NIR) imaging camera for secure authentication.
GSM/GPRS Module	SIM800L / SIM7600 (4G LTE)	Handles remote telemetry, SMS triggers, and automated voice dialing.
GPS Module	Neo-6M / Neo-M8N	High-sensitivity navigation module with active patch antenna.
Inertial Sensor	MPU6050 / ADXL345	3-axis accelerometer and gyroscope for crash vector analysis.
Display Unit	16x2 I2C LCD / 3.5" TFT Touch	Real-time diagnostic messages, countdown timers, and checklist UI.
Relay Module	5V Optocoupler Isolated Relay	Safely interrupts the truck's 12V/24V starter ignition circuit.
Diagnostic Sensors	TPMS Sensors, Load cells, DS18B2V	Real-time feedback loops for the pre-drive checklist.

5.2 Software Stack & Protocols

- **Embedded Firmware:** Written in C/C++ using the FreeRTOS framework on ESP32 to manage multi-threaded task scheduling (e.g., polling sensors while keeping GSM active).
- **Mobile App Framework:** Developed using Flutter/React Native for cross-platform (Android/iOS) BLE background scanning, call blocking permission handlers, and automated SMS broadcasts.
- **Backend & Cloud Database:** Node.js/Express server linked to a MongoDB database to aggregate trip logs, diagnostic parameters, driver check-in times, and incident telemetry.

6. Business & Safety Benefits

Implementing Navisecure offers profound advantages across multiple stakeholders

- **For Fleet Owners:**
 - *Reduced Asset Loss:* Eliminates unauthorized runs or vehicle theft via robust biometric checks.
 - *Predictive Maintenance:* Continuous collection of TPMS and load metrics reduces breakdown risks and extends tire life.
 - *Regulatory Compliance:* Ensures trucks do not exceed local road loading regulations.
- **For Drivers:**
 - *Mitigates Highway Distraction:* Acts as a physical boundary against mobile phone interruptions, keeping drivers focused.
 - *Survival Guarantee:* Ensures that if an incapacitating crash occurs, help is automatically dispatched to the precise coordinates immediately, eliminating delays.
- **For Insurance and Forensic Investigators:**
 - *Accident Reconstruction:* The dedicated Black Box unit provides empirical telemetry data (speed, steering corrections, impact vector) to clarify liability and settle insurance disputes.

7. Conclusion & Future Roadmaps

The Navisecure prototype bridges a long-standing technology gap in the commercial shipping and heavy transportation sector. By merging biometric access, localized smart device integration, real-time sensor monitoring, and automated emergency telemetry, the system successfully moves vehicle safety from passive containment to connected protection.

Future Expansion Milestones

- **Computer Vision Fatigue Tracking (DMS):** Upgrading the cabin camera processing

pipeline to monitor driver yawn rate, blink durations, and head posture, generating warnings before micro-sleep occurs.

- **Advanced Driver Assistance Systems (ADAS):** Integrating low-cost LiDAR or radar sensors for forward-collision warnings and lane departure alerts.
- **AI Engine Diagnosis:** Applying machine learning on the collected cloud database to predict mechanical issues or engine degradation before physical warning lights illuminate.